

Notes: Denis and Sibilkov (RFS, 2010)

Financial Constraints, Investment, and the Value of Cash Holdings

One-sentence summary. Denis and Sibilkov (2010) show that cash holdings are more valuable for financially constrained firms because cash helps these firms finance value-increasing investment, especially when cash flows are poorly aligned with investment opportunities; constrained firms that hold little cash appear cash-poor mainly because weak and declining internal funds prevent them from building reserves, not because they face fewer investment needs or stronger agency discipline.

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1 Big picture

1.1 Main question

The paper asks two connected questions. First, why do prior studies find that an additional dollar of cash is more valuable in financially constrained firms than in unconstrained firms? Second, if cash is especially valuable for constrained firms, why do some constrained firms hold very little cash?

The paper’s answer is fundamentally about financing frictions. When external finance is costly, limited, or unavailable, internal liquidity becomes a real productive resource. Cash is valuable not simply because investors like liquid balance sheets, but because cash allows constrained firms to fund projects that would otherwise be delayed or abandoned.

1.2 Economic mechanism

The paper’s mechanism has three parts:

1. **External finance is costly for constrained firms.** These firms cannot always raise capital on favorable terms when investment opportunities arrive.
2. **Cash relaxes the constraint.** Cash holdings allow constrained firms to invest even when current cash flows are low or poorly matched with investment needs.
3. **The marginal investment is valuable.** The investment undertaken by constrained firms is more positively related to firm value than investment by unconstrained firms, suggesting that the marginal projects made possible by cash are valuable.

1.3 High-level conclusion

The evidence supports the value-increasing cash-retention view rather than the cash-hoarding agency view. Higher cash balances in constrained firms are not mainly interpreted as wasteful cash piles that enable empire building. Instead, they are a response to costly external financing. However, some constrained firms have little cash because their weak operating performance and persistently negative free cash flow leave them unable to save.

2 Incremental contribution of the paper

2.1 Moves from “cash is valuable” to “why cash is valuable”

Prior work documents that cash is more valuable in constrained firms. This paper pushes further and tests the channel. It links cash to investment and investment to market value. This matters because a high marginal value of cash could come from many sources. Denis and Sibilkov show that the investment channel is central.

2.2 Uses several constraint classifications

The paper avoids relying on only one noisy measure of financial constraints. It uses four common classifications: payout ratio, firm size, bond ratings, and commercial paper ratings. The fact that the main results appear across these alternative classifications strengthens the interpretation that the findings are about financial constraints rather than one mechanical sorting rule.

2.3 Introduces hedging needs into the cash-investment link

The paper borrows the logic of Acharya, Almeida, and Campello: cash is especially useful when cash flows are low exactly when investment opportunities are high. Firms with this pattern have high hedging needs. Denis and Sibilkov show that the cash-investment relation is concentrated among constrained firms with high hedging needs. This is one of the strongest pieces of evidence that cash is actually relaxing investment constraints.

2.4 Explains why low-cash constrained firms exist

A simple theory might say all constrained firms should hold lots of cash. The paper shows why that need not happen. Some constrained firms are cash-poor because they have persistently low cash flows and negative free cash flow. They cannot build reserves; when cash flow arrives, it is used for investment or operating needs rather than saved.

3 Data, sample, and classification

3.1 Sample

The sample consists of U.S. public firms from Compustat over 1985–2006. The main sample contains 74,347 firm-year observations after excluding financial firms, utilities, very small firms, and firm-years with nonpositive assets or cash holdings or negative capital expenditures.

3.2 Measures of financial constraints

The paper uses four ways to classify firms as financially constrained or unconstrained:

1. **Payout ratio:** low payout firms are classified as constrained; high payout firms as unconstrained.
2. **Size:** small firms are classified as constrained; large firms as unconstrained.
3. **Bond ratings:** firms with rated public debt are treated as unconstrained; firms with debt but no rating history are treated as constrained.
4. **Commercial paper ratings:** firms with rated short-term debt are treated as unconstrained; firms with short-term debt but no rating are treated as constrained.

The classifications are correlated but not identical, which is useful because each proxy captures a different dimension of financing access.

3.3 Main variables

Cash holdings, net investment, capital expenditures, cash flow, debt, property, plant, and equipment, and related balance sheet variables are generally scaled by total assets. Net investment is capital expenditures net of depreciation. Firm value tests use abnormal stock returns, following the Faulkender and Wang methodology, to infer the market value of marginal cash and investment.

4 Empirical strategy

4.1 Testing whether cash is more valuable for constrained firms

The first test estimates whether annual excess stock returns respond more strongly to changes in cash for constrained firms. The key coefficient is the interaction between the change in cash and the constrained-firm indicator. A positive coefficient means the market values an additional dollar of cash more in constrained firms.

4.2 Testing whether cash increases investment

The second test asks whether firms with more cash invest more. Because cash and investment may be jointly determined by investment opportunities, the paper estimates a three-stage least squares system with a cash equation and an investment equation. The investment equation includes cash holdings, cash flow, market-to-book, prior sales growth, and fixed effects.

4.3 Testing the hedging-needs channel

The third test splits constrained firms into high- and low-hedging-needs groups. High hedging needs means the firm's cash flows are negatively related to industry investment opportunities. If cash truly relaxes constraints, then the cash-investment relation should be strongest exactly for these firms.

4.4 Testing whether constrained-firm investment is more valuable

The fourth test augments the excess-return regressions with an interaction between changes in noncash assets and the constrained-firm indicator. The idea is that changes in noncash assets proxy for investment. A positive interaction means that the market responds more favorably to investment by constrained firms.

4.5 Explaining low cash among constrained firms

The final tests compare low-cash and high-cash constrained firms. The paper examines managerial ownership, investment, financial health, external issuance, analyst coverage, past cash flows, free cash flows, and the sensitivities of cash and investment to cash flow.

5 Main findings

5.1 Constrained firms hold more cash but are financially weaker

Across the four financial constraint criteria, constrained firms hold more cash than unconstrained firms and exhibit more volatile cash balances. At the same time, they tend to have lower cash flow, smaller size, fewer tangible assets, and slightly lower market-to-book ratios. This is consistent with the idea that constrained firms save liquidity because outside financing is costly.

5.2 Cash is more valuable for constrained firms

The excess-return tests show that an increase in cash is positively associated with abnormal stock returns, and the association is stronger for constrained firms. Depending on the classification, the incremental value of cash for constrained firms is about 14 to 51 cents higher than for unconstrained firms.

5.3 Cash supports investment, especially when hedging needs are high

The baseline cash-investment regressions show a positive relation between cash and investment for constrained firms, although the comparison with unconstrained firms is mixed. The cleaner evidence comes from the high-hedging-needs split: cash holdings strongly predict investment for constrained firms with high hedging needs, while the coefficient is insignificant for constrained firms with low hedging needs.

5.4 Investment by constrained firms is more valuable

Changes in noncash assets are positively related to excess returns, and this relation is stronger for constrained firms. This supports the interpretation that the projects funded by constrained firms are high-value marginal projects.

5.5 Low-cash constrained firms are cash-poor, not necessarily over-disciplined

The low-cash constrained firms do not exhibit clear signs of stronger agency discipline or lower investment needs. Their CEO ownership and insider ownership look similar to high-cash constrained firms, and their net investment is not higher. Instead, they have worse financial health: lower cash flow, lower Altman's z -scores, lower interest coverage, declining margins, lower analyst coverage, and less external capital issuance.

6 Table-by-table and figure-by-figure interpretation

6.1 Table 1: correlation among financial constraint classifications

Table 1 reports correlations among the four financial constraint proxies. The correlations are all positive and statistically significant, ranging from 0.1835 to 0.6582. The highest correlation is between size and bond ratings, which makes sense because larger firms are more likely to have public debt ratings. The lower correlations show that the proxies are not interchangeable. This matters because the paper's main results are stronger when they survive across several imperfect measures.

6.2 Table 2: univariate characteristics of constrained and unconstrained firms

Table 2 shows that constrained firms generally hold more cash. For example, under the size criterion, mean cash holdings are 21.1% of assets for constrained firms and 8.7% for unconstrained firms. Under the bond-rating criterion, mean cash holdings are 17.7% versus 8.6%. The table also shows that constrained firms have lower cash flow and are much smaller. This combination is important: constrained firms hold more cash not because they are flush with internal funds, but because precautionary cash is more valuable when external funds are costly.

6.3 Table 3: excess stock returns and changes in cash holdings

Table 3 replicates a marginal-value-of-cash style regression. The coefficient on the change in cash is positive and significant in every specification. More importantly, the interaction between the constrained dummy and the change in cash is positive and significant across all four constraint definitions. The estimates imply that the marginal value of cash is higher in constrained firms. The interpretation is that investors value cash more when the firm is likely to face costly external finance.

6.4 Table 4: determinants of net investment

Table 4 estimates the relation between cash and subsequent net investment using a 3SLS system. Cash is positively related to investment for constrained firms under all four classification methods, with cash coefficients ranging from 0.032 to 0.053. For unconstrained firms, the relation is more uneven. The paper describes this baseline evidence as mixed because the difference between constrained and unconstrained coefficients is not uniformly significant. Still, the table establishes the first step of the investment channel: cash and investment are positively related among constrained firms.

6.5 Table 5: constrained firms with high and low hedging needs

Table 5 is one of the paper’s most important tables. For constrained firms with high hedging needs, cash strongly predicts investment across all four constraint definitions. The cash coefficients are 0.117, 0.065, 0.103, and 0.094, all statistically significant. For constrained firms with low hedging needs, the cash coefficients are economically small and statistically insignificant. The p-values for differences are 0.00, 0.01, 0.00, and 0.00. This is powerful because it shows that cash matters precisely when external financing frictions should make liquidity most valuable.

6.6 Table 6: investment and excess returns

Table 6 asks whether investment is more valuable in constrained firms. The interaction between constrained status and changes in noncash assets is positive and significant in all four specifications. For example, the coefficient is 0.156 under the size criterion and 0.139 under the commercial paper ratings criterion. This suggests that the market responds more favorably to investment by constrained firms. Combined with Tables 4 and 5, the evidence supports a complete chain: cash helps constrained firms invest, and that investment is valuable.

6.7 Table 7: low-cash versus high-cash constrained firms

Table 7 investigates why some constrained firms hold little cash. The agency interpretation would predict that low-cash constrained firms might have stronger governance or more disciplined managers. The table does not support this: CEO and insider ownership are broadly similar across low- and high-cash constrained firms. Instead, low-cash constrained firms look financially weaker. They have lower cash flow, lower z -scores, lower interest coverage, weaker margin growth, lower debt and equity issuance, and lower analyst coverage. The most natural interpretation is that these firms are unable to accumulate cash, not that they optimally choose low cash because cash is unimportant.

6.8 Figure 1: characteristics of low-cash constrained firms over time

Figure 1 plots low-cash constrained firms over the ten years before the classification year. Mean cash flow declines sharply, from roughly 8% of assets about ten years earlier to about 1.5% by year 0. Free cash flow is negative throughout and becomes more negative over time. Net investment and cash holdings are relatively stable earlier but fall in the final years. The figure gives the time-series version of Table 7’s message: these firms appear trapped by weak internal funds and persistent negative free cash flow.

6.9 Table 8: cash-flow sensitivity of cash and investment

Table 8 formalizes the low-cash-firm mechanism. Panel A shows that low-cash constrained firms generally have a lower cash-flow sensitivity of cash than high-cash constrained firms. That means additional cash flow is less likely to be saved as cash. Panel B shows that low-cash firms have higher investment-cash flow sensitivity than high-cash firms. This pattern suggests that low-cash constrained firms use available cash flow to fund investment and cannot build liquidity buffers. It supports the interpretation that their low cash balances reflect financial weakness rather than easy access to alternative liquidity.

7 How the results fit together

7.1 The paper’s causal chain

The paper’s main interpretation can be summarized as follows:

financial constraints $\Rightarrow$ cash is valuable $\Rightarrow$ cash supports investment $\Rightarrow$ investment raises value.

The strongest evidence for the middle link is Table 5, because it isolates firms for whom cash should be most useful as a hedge against mismatch between cash flows and investment opportunities.

7.2 Why Table 5 matters more than Table 4

Table 4 establishes that cash and investment are positively related for constrained firms, but the comparison with unconstrained firms is not uniformly sharp. Table 5 provides a cleaner economic test. If cash were merely a proxy for managers’ private information about opportunities, there is no strong reason for the cash-investment coefficient to be concentrated only in high-hedging-needs firms. The fact that it is concentrated there supports the financing-frictions interpretation.

7.3 Why low cash is not a contradiction

Low cash among constrained firms is initially puzzling. If cash is valuable, why not hold more? The answer is that desired cash and feasible cash are different. The low-cash constrained firms have weak cash flows and negative free cash flows. They may want more cash but cannot accumulate it because available funds are absorbed by current operations and investment needs.

8 Interpretation, caveats, and limitations

8.1 Endogeneity remains difficult

The authors use 3SLS to address simultaneity between cash and investment, but they are careful not to claim that all endogeneity is solved. Cash can still contain information about investment opportunities, future profitability, or managerial expectations. The high-hedging-needs split helps, but does not fully eliminate all identification concerns.

8.2 Constraint classifications are noisy

Payout, size, bond ratings, and paper ratings are proxies for financial constraints. They capture related but different things. Some small firms are not constrained, some large firms are constrained, and debt ratings may reflect both financing access and firm maturity. The use of several proxies is a strength, but noisy classification can attenuate or complicate estimates.

8.3 Market valuation tests are associative

The excess-return regressions show how stock returns are associated with changes in cash and investment. These regressions are informative about investor valuation, but they are not a randomized experiment. A positive relation between investment and returns may partly reflect news about future profitability.

8.4 The paper is about corporate cash as productive liquidity

The paper does not deny that cash can generate agency costs in some firms. Its point is more specific: among financially constrained firms, higher cash holdings appear to serve a valuable financing role, and the evidence does not support the view that constrained firms' higher cash balances are mainly wasteful hoarding.

9 Takeaways

1. Cash is more valuable in financially constrained firms because it helps finance investment.
2. The investment channel is strongest when firms have high hedging needs, meaning cash flows are poorly timed relative to investment opportunities.
3. Investment by constrained firms is more strongly associated with value than investment by unconstrained firms.
4. Low-cash constrained firms seem unable to accumulate cash because of weak and declining internal funds.
5. The findings support the view that cash retention can be an efficient response to financing frictions.

10 Extracted tables and figure from the paper

Table 1
Correlation of various classifications of financial constraints

	Financial constraint criteria		
	Payout ratio	Size	Bond ratings
Payout ratio	1		
(<i>p</i> -value)			
Size	0.3822	1	
(<i>p</i> -value)			
Bond ratings	0.1835	0.6582	1
(<i>p</i> -value)			
Paper ratings	0.2893	0.4277	0.4446
(<i>p</i> -value)			(0.00)

Data are from Compustat for the period of 1985–2006. We exclude companies from the financial (SIC 6000–6999) and utility (SIC 4910–4939) industries. We also exclude firms with total book assets less than twenty-five million in constant 1994 dollars and firm-years with nonpositive assets or cash holdings, or negative capital expenditures. See the text for definitions of criteria used to categorize firm-years as financially constrained or unconstrained. Correlation coefficient estimates are reported with *p*-values in parentheses below.

Table 1. Correlation of financial constraint classifications. The four constraint proxies are positively related but far from identical, motivating the use of multiple classifications.

Table 2
Univariate comparison by financial constraints

		Financial constraint criteria							
		Payout		Size		Bond ratings		Paper ratings	
		Uncon. (A)	Constr. (B)	Uncon. (A)	Constr. (B)	Uncon. (A)	Constr. (B)	Uncon. (A)	Constr. (B)
Cash holdings	Mean	0.118	0.174	0.087	0.211	0.086	0.177	0.067	0.160
	Median	0.055	0.085	0.046	0.112	0.042	0.085	0.037	0.073
St. dev. cash	Mean	0.068	0.089	0.056	0.095	0.060	0.084	0.046	0.080
	Median	0.053	0.078	0.044	0.085	0.046	0.070	0.039	0.066
St. dev. cash chg.	Mean	0.083	0.131	0.067	0.134	0.076	0.117	0.045	0.111
	Median	0.054	0.091	0.045	0.094	0.049	0.077	0.037	0.072
Cash flow	Mean	0.067	0.027	0.080	0.019	0.065	0.042	0.101	0.044
	Median	0.085	0.064	0.087	0.066	0.076	0.078	0.102	0.074
Net investment	Mean	0.015	0.023	0.024	0.030	0.022	0.030	0.020	0.028
	Median	0.004	0.001	0.010	0.003	0.006	0.007	0.011	0.006
Size	Mean	3180	756	5675	65	4207	811	9407	1018
	Median	510	161	2126	54	1219	122	4933	176
Total debt	Mean	0.275	0.276	0.311	0.214	0.387	0.232	0.283	0.275
	Median	0.245	0.227	0.281	0.142	0.340	0.182	0.268	0.232
PPE	Mean	0.353	0.293	0.396	0.268	0.377	0.304	0.401	0.317
	Median	0.292	0.224	0.352	0.195	0.330	0.240	0.353	0.254
Market-to-book	Mean	2.914	2.758	2.926	2.751	2.934	2.782	3.275	2.776
	Median	1.991	1.819	2.050	1.795	2.019	1.848	2.354	1.847

Data are from Compustat for the period of 1985–2006. We exclude companies from the financial (SIC 6000–6999) and utility (SIC 4910–4939) industries. We also exclude firms with total book assets less than twenty-five million in constant 1994 dollars and firm-years with nonpositive assets or cash holdings, or negative capital expenditures. See the text for definitions of financial constraint criteria. Size is total book assets measured in 1994 dollars (adjusted for the CPI). Cash holdings, property, plant, and equipment, total debt, net investment, and cash flow are deflated by total book assets.

Table 2. Univariate comparison by financial constraints. Constrained firms hold more cash and have weaker cash flow, supporting the precautionary liquidity interpretation.

Table 3
Excess stock returns and changes in cash holdings

	Financial constraint criteria			
	Payout ratio	Size	Bond ratings	Paper ratings
Change in cash	0.953*** (8.77)	0.964*** (7.58)	1.017*** (10.40)	0.629*** (4.30)
Constrained*change in cash	0.267*** (2.91)	0.287*** (3.05)	0.137** (2.24)	0.511*** (3.86)
Change in noncash assets	0.180*** (13.76)	0.188*** (11.01)	0.187*** (13.37)	0.187*** (13.39)
Change in earnings	0.333*** (13.36)	0.374*** (11.03)	0.362*** (14.85)	0.361*** (14.79)
Change in R&D	0.677*** (3.74)	0.729*** (3.32)	0.841*** (5.24)	0.841*** (5.33)
Change in interest expense	-1.473*** (8.96)	-1.628*** (6.81)	-1.672*** (10.70)	-1.682*** (10.73)
Change in dividends	3.406*** (10.62)	3.333*** (7.22)	3.618*** (14.05)	3.583*** (14.01)
Net financing	-0.047* (1.82)	-0.055* (1.80)	-0.039 (1.66)	-0.039 (1.67)
Leverage	-0.005 (0.95)	-0.006 (1.11)	-0.005 (1.19)	-0.005 (1.96)
Lagged cash	0.278*** (9.87)	0.265*** (7.97)	0.261*** (10.32)	0.268*** (10.39)
Change in cash*lagged cash	-0.300*** (3.13)	-0.371*** (2.69)	-0.288*** (2.98)	-0.281*** (2.93)
Change in cash*leverage	-0.045*** (2.97)	-0.037 (1.60)	-0.048*** (3.67)	-0.053*** (3.91)
Constrained dummy	0.043*** (4.49)	-0.023*** (3.15)	-0.013** (2.05)	-0.037*** (5.05)
Observations	23,094	17,109	31,366	31,366
Adjusted R ²	0.158	0.154	0.155	0.156

Data are from Compustat for the period of 1985–2006. We exclude companies from the financial (SIC 6000–6999) and utility (SIC 4910–4939) industries. We also exclude firms with total book assets less than twenty-five million in constant 1994 dollars and firm-years with nonpositive assets or cash holdings, or negative capital expenditures. See the text for definitions of financial constraint criteria. The dependent variable is stock return over fiscal year minus the return on a benchmark portfolio. The benchmark portfolios are twenty-five French value-weighted portfolios. The independent variables include change in cash, change in book assets net of cash, change in earnings before interest and extraordinary items, change in R&D expenses, change in interest expenses, change in dividends, lagged cash holdings, leverage, and net financing during fiscal year. All explanatory variables except leverage are standardized by lagged market equity. The FC dummy takes a value of 1 if a company is identified as financially constrained and 0 if a company is identified as financially unconstrained by the respective criterion. Regressions are estimated using OLS. Statistical significance is computed using heteroscedasticity and autocorrelation robust standard errors. ***, **, and * denote significance at the 1, 5, and 10% levels, respectively. Coefficient estimates are reported with *t*-statistics in parentheses below.

Table 3. Excess stock returns and changes in cash holdings. The constrained-firm interaction with cash is positive across all criteria, showing a higher marginal value of cash in constrained firms.

Table 4
Determinants of net investment

Financial constraint criteria	Payout ratio	Size	Bond ratings	Paper ratings
Constrained firms				
Cash	0.053*** (7.22)	0.044*** (5.45)	0.032*** (5.92)	0.041*** (7.72)
Cash flow	0.084*** (20.54)	0.081*** (17.04)	0.080*** (24.92)	0.077*** (25.61)
Market-to-book	0.007*** (12.16)	0.006*** (9.23)	0.007*** (16.36)	0.007*** (17.01)
Prior sales growth	0.006*** (13.64)	0.004*** (6.42)	0.005*** (14.29)	0.006*** (17.67)
Observations	21372	10279	31297	43192
Unconstrained firms				
Cash	0.021*** (2.83)	0.036** (2.44)	-0.002 (0.18)	0.091*** (3.25)
Cash flow	0.055*** (12.46)	0.061*** (10.51)	0.045*** (7.45)	0.073*** (9.03)
Market-to-book	0.005*** (9.55)	0.006*** (8.72)	0.008*** (12.34)	0.004*** (4.22)
Prior sales growth	0.002*** (4.32)	0.006*** (12.32)	0.006*** (13.02)	0.007*** (5.70)
Observations	15,348	18,481	19,093	7198
<i>p</i> -value for difference in the coefficients on cash	0.00	0.63	0.00	0.08

Data are from Compustat for the period of 1985–2006. We exclude companies from the financial (SIC 6000–6999) and utility (SIC 4910–4939) industries. We also exclude firms with total book assets less than twenty-five million in constant 1994 dollars and firm-years with nonpositive assets or cash holdings, or negative capital expenditures. See the text for definitions of financial constraint criteria. The dependent variable is net investment, defined as capital expenditures net of depreciation in year $(t + 1)$. Net investment, cash, and cash flow are deflated by total book assets. Prior growth is measured as growth in sales for the past two years. Regressions are estimated using the 3SLS methodology, and include industry fixed effects and calendar-year dummies. Industry is defined at the two-digit SIC level. The unreported cash holdings model has cash-to-total assets as the dependent variable and firm size, market-to-book ratio, leverage, prior twenty-year industry cash flow volatility, duration of cash cycle, operating cash flow, likelihood of financial distress, return spread between a firm's investments and the return on Treasury bills, and log of the growth rate of industrial production as control variables. The structure of standard errors allows for unstructured correlation across models. ***, **, and * denote significance at the 1, 5, and 10% levels, respectively.

Table 4. Determinants of net investment. Cash is positively associated with investment for constrained firms, although the constrained-versus-unconstrained contrast is not uniformly sharp.

Table 5
Determinants of net investment for constrained firms with high and low hedging needs

Hedging needs	Correlation between cash flow and industry sales growth			
	Payout ratio	Size	Bond ratings	Paper ratings
High hedging needs				
Cash	0.117*** (5.06)	0.065** (2.47)	0.103*** (5.98)	0.094*** (5.64)
Observations	2014	965	3392	4847
Low hedging needs				
Cash	0.023 (1.05)	-0.031 (1.24)	0.006 (0.38)	-0.007 (0.45)
Observations	2260	1057	3259	4609
p-value for difference	0.00	0.01	0.00	0.00

Data are from Compustat for the period of 1985–2006. We exclude companies from the financial (SIC 6000–6999) and utility (SIC 4910–4939) industries. We also exclude firms with total book assets less than twenty-five million in constant 1994 dollars and firm-years with nonpositive assets or cash holdings, or negative capital expenditures. See the text for definitions of financial constraint criteria. Firms with the statistically significant correlation of -0.2 or less between firm cash flow and investment opportunities are classified as having high hedging needs, and those with the statistically significant correlation of 0.2 or more as having low hedging needs. Three-digit SIC industry median sales growth over the next three years and next year's industry median net investment proxy for the firm's investment opportunities. The dependent variable is net investment, defined as capital expenditures net of depreciation in year $(t + 1)$. Net investment, cash, and cash flow are deflated by total book assets. Prior growth is measured as growth in sales for the past two years. Regressions are estimated using the 3SLS methodology, and include industry fixed effects and calendar-year dummies. Industry is defined at the two-digit SIC level. The unreported cash holdings model has cash-to-total assets as the dependent variable and firm size, market-to-book ratio, leverage, prior twenty-year industry cash flow volatility, duration of cash cycle, operating cash flow, likelihood of financial distress, return spread between a firm's investments and the return on Treasury bills, and log of the growth rate of industrial production as control variables. The structure of standard errors allows for unstructured correlation across models. *** and ** denote significance at the 1, 5, and 10%

Table 5. High versus low hedging needs. Cash matters for constrained firms only when hedging needs are high, which is central evidence for the liquidity-relaxation mechanism.

Table 6
Excess stock returns and investment

	Financial constraint criteria			
	Payout ratio	Size	Bond ratings	Paper ratings
Change in cash	0.942*** (8.43)	0.960*** (7.13)	1.014*** (10.10)	0.561*** (4.15)
Constrained*change in cash	0.280*** (3.03)	0.306*** (3.11)	0.142** (2.42)	0.582*** (4.79)
Change in noncash assets	0.148*** (11.30)	0.105*** (7.10)	0.133*** (9.83)	0.056** (2.12)
Constrained*change in noncash assets	0.045*** (2.88)	0.156*** (7.84)	0.083*** (5.55)	0.139*** (4.87)
Change in earnings	0.335*** (13.36)	0.384*** (11.70)	0.363*** (15.26)	0.359*** (14.57)
Change in R&D	0.676*** (3.69)	0.697*** (3.40)	0.817*** (5.14)	0.835*** (5.46)
Change in interest expense	-1.466*** (8.97)	-1.645*** (6.80)	-1.659*** (10.55)	-1.683*** (10.93)
Change in dividends	3.444*** (10.67)	3.741*** (8.04)	3.708*** (14.42)	3.800*** (14.59)
Net financing	-0.049* (1.87)	-0.066** (2.09)	-0.032 (1.33)	-0.042* (1.76)
Leverage	-0.004 (0.81)	-0.010* (1.75)	-0.007 (1.47)	-0.004 (0.96)
Lagged cash	0.276*** (9.76)	0.259*** (7.90)	0.259*** (10.37)	0.267*** (10.34)
Change in cash*lagged cash	-0.305*** (3.19)	-0.380*** (2.76)	-0.292*** (3.05)	-0.287*** (2.99)
Change in cash*leverage	-0.046*** (3.00)	-0.035 (1.60)	-0.048*** (3.62)	-0.052*** (3.81)
Constrained dummy	0.044*** (4.51)	-0.025*** (3.51)	-0.015** (2.33)	-0.041*** (5.91)
Observations	23,094	17,109	31,366	31,366
Adjusted R^2	0.158	0.160	0.157	0.157

Data are from Compustat for the period of 1985–2006. We exclude companies from the financial (SIC 6000–6999) and utility (SIC 4910–4939) industries. We also exclude firms with total book assets less than twenty-five million in constant 1994 dollars and firm-years with nonpositive assets or cash holdings, or negative capital expenditures. See the text for definitions of financial constraint criteria. The dependent variable is stock return over fiscal year minus the return on a benchmark portfolio. The benchmark portfolios are twenty-five Fama-French value-weighted portfolios. The independent variables include change in cash, change in book assets net of cash, change in earnings before interest and extraordinary items, change in R&D expenses, change in interest expenses, change in dividends, lagged cash holdings, leverage, and net financing during fiscal year. All explanatory variables except leverage are standardized by lagged market equity. The FC dummy takes a value of 1 if a company is identified as financially constrained and 0 if a company is identified as financially unconstrained

Table 6. Excess stock returns and investment. The positive constrained-firm interaction with noncash assets indicates that investment by constrained firms is more strongly valued by the market.

Table 7
Univariate comparison of financially constrained firms

		Financial constraint criteria							
		Payout		Size		Bond ratings		Paper ratings	
		Low (A)	High (B)	Low (A)	High (B)	Low (A)	High (B)	Low (A)	High (B)
Cash									
CEO ownership	Mean	0.056	0.051	0.079	0.073	0.059	0.063*	0.052	0.054*
Insider ownership	Mean	0.096	0.088	0.130	0.125	0.101	0.105	0.089	0.092*
Net investment	Mean	0.024	0.025	0.027	0.035*	0.029	0.033*	0.028	0.031*
Cash flow	Mean	0.022	0.032*	0.012	0.026*	0.037	0.051*	0.040	0.050*
z-score	Median	3.362	3.758*	3.719	6.050*	3.994	5.460*	3.596	3.949*
Interest coverage	Mean	11.863	24.654*	11.663	26.886*	15.376	30.487*	13.590	24.605*
Three-year change in margin	Mean	-0.002	0.014*	0.000	0.021*	-0.004	0.008*	-0.003	0.008*
Value of debt issued	Mean	0.082	0.108*	0.103	0.163*	0.079	0.103*	0.078	0.093*
Value of equity issued	Mean	0.053	0.062*	0.056	0.064*	0.049	0.051*	0.055	0.061*
Analyst coverage	Mean	56.0	64.2*	22.9	28.8*	42.6	46.5*	53.1	57.2*

Data are from Compustat for the period of 1985–2006. We exclude companies from the financial (SIC 6000–6999) and utility (SIC 4910–4939) industries. We also exclude firms with total book assets less than twenty-five million in constant 1994 dollars and firm-years with nonpositive assets or cash holdings, or negative capital expenditures. See the text for definitions of financial constraint criteria. The sample includes financially constrained firms only. Firms with the ratio of abnormal cash holdings to total assets below (above) the median are classified as low (high) cash firms. Cash flow and net investment are deflated by total book assets. CEO ownership and insider ownership are the percentage of equity (common stock + options) held by the top executive officer and by top executive officers, respectively, as reported by S&P's Execucomp database. Analyst coverage is the number of annual analyst recommendations for a firm. The value of debt issues is long-term debt issuance net of reduction over total book assets. The value of equity issues is one-year change in the number of stocks outstanding adjusted for stock splits multiplied by the fiscal year close price. Three-year change in margin is the change in cash flow margin over the prior three-year period. Cash flow margin is income before extraordinary items plus depreciation and amortization over sales. z-score is Altman's (1968) z-score. Interest coverage is earnings before interest and tax divided by interest expense. Mean values of each variable are reported, except for z-score where we report the median due to extreme observations. * denotes that the value for high cash firms is significantly different

Table 7. Low-cash and high-cash constrained firms. Low-cash constrained firms look financially weaker rather than better governed, pointing to an inability to accumulate liquidity.

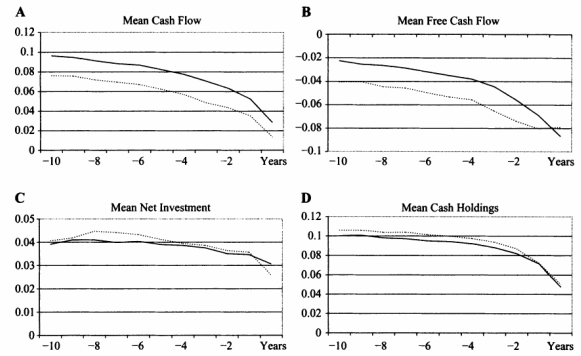


Figure 1
Characteristics of low cash constrained firms
Data are from Compustat for the period of 1985–2006. We exclude companies that belong to the financial (SIC 6000–6999) and utility (SIC 4910–4939) industries. We exclude firms with total book assets less than twenty-five million in constant 1994 dollars and firm-years with nonpositive assets or cash holdings, or negative capital expenditures. Values reported are for low cash constrained firms. Dashed lines represent payout constraint criterion and solid lines represent bond rating constraint criterion. See the text for definitions of financial constraint criteria. Cash flow, free cash flow, net investment, and cash holdings are deflated by contemporaneous total book assets. Mean values of each variable are reported.

Figure 1. Characteristics of low-cash constrained firms. Cash flow falls, free cash flow remains negative, and cash holdings decline, consistent with persistent internal-funds shortages.

Table 8
Determinants of changes in cash and net investment

	Payout			Size			Bond ratings			Paper ratings		
	Low	High	p-value	Low	High	p-value	Low	High	p-value	Low	High	p-value
Cash												
Net cash flow	0.103*** (3.22)	0.263*** (7.24)	0.00	0.152*** (4.97)	0.262*** (4.71)	0.08	0.148*** (4.70)	0.324*** (10.04)	0.00	0.112*** (5.14)	0.242*** (5.76)	0.01
Capex	0.023 (0.98)	-0.035 (1.03)	0.16	0.038 (1.15)	-0.020 (0.31)	0.42	-0.010 (0.57)	-0.078*** (2.95)	0.04	0.006 (0.41)	-0.045* (1.94)	0.07
Acquisitions	-0.018 (1.17)	-0.263*** (8.28)	0.00	-0.014 (0.56)	-0.242*** (5.43)	0.00	-0.010 (0.87)	-0.238*** (10.69)	0.00	-0.009 (1.29)	-0.212*** (11.94)	0.00
NWC change	-0.067*** (3.38)	-0.169*** (6.02)	0.00	-0.102*** (4.44)	-0.245*** (5.77)	0.00	-0.103*** (8.07)	-0.234*** (10.52)	0.00	-0.085*** (6.85)	-0.188*** (9.21)	0.00
Adj. R ²	2.37%	7.31%		3.52%	7.45%		3.86%	9.80%		2.86%	7.79%	
Observations	11,593	11,786		6,549	5,883		17,589	17,476		23,307	22,996	
Cash flow	0.105*** (16.06)	0.085*** (12.29)	0.04	0.100*** (14.07)	0.081*** (9.05)	0.10	0.099*** (20.00)	0.083*** (15.16)	0.03	0.100*** (21.93)	0.076*** (15.71)	0.00
Cash holdings	0.087*** (3.80)	0.083*** (7.83)	0.87	0.093*** (3.77)	0.078*** (5.64)	0.60	0.061*** (3.64)	0.061*** (7.55)	1.00	0.094*** (5.99)	0.064*** (8.98)	0.08
Market-to-book	0.009*** (9.92)	0.005*** (7.21)	0.00	0.009*** (7.84)	0.005*** (4.79)	0.01	0.009*** (13.64)	0.005*** (8.78)	0.00	0.008*** (13.74)	0.005*** (10.48)	0.00
Lag (sales growth)	0.006*** (8.51)	0.006*** (10.57)	1.00	0.006*** (5.55)	0.003*** (3.50)	0.03	0.004*** (7.84)	0.006*** (12.10)	0.00	0.005*** (9.87)	0.006*** (14.88)	0.12
Observations	10,655	9,865		5,470	4,277		15,879	14,207		21,864	19,944	

Data are from Compustat for the period of 1985–2006. We exclude companies from the financial (SIC 6000–6999) and utility (SIC 4910–4939) industries. We also exclude firms with total book assets less than twenty-five million in constant 1994 dollars and firm-years with nonpositive assets or cash holdings, or negative capital expenditures. See the text for definitions of financial constraint criteria. The dependent variable in panel A is the change in cash holdings. The dependent variable in panel B is net investment, defined as capital expenditures net of depreciation in year $t + 1$. The models are estimated separately for low cash and high cash constrained firms, where low (high) cash firms are defined as those with abnormal cash below (above) the median. Cash flow, capital expenditures, acquisitions, net working capital change, and cash holdings are deflated by total book assets. Prior growth is measured as growth in sales for the past two years. Regressions in panel A are estimated using firm fixed effects and clustered standard errors. Regressions in panel B are estimated using the 3SLS methodology, and include industry fixed effects and calendar-year dummies. Industry is defined as the two-digit SIC level. The unreported cash holdings model in panel B has cash-to-total assets as the dependent variable and firm size, market-to-book ratio, leverage, prior twenty-year industry cash flow volatility, duration of cash cycle, operating cash flow, likelihood of financial distress, return spread between a firm's investments and the return on Treasury bills, and log of the growth rate of industrial production as control variables. The structure of standard errors in panel B allows for unstructured correlation across models. *** and * denote significance at the 1, 5, and 10% levels, respectively. Coefficient estimates are reported with t -statistics (z -statistics) in parentheses in panel A (panel B) below.

Table 8. Determinants of changes in cash and net investment. Low-cash constrained firms save less of cash-flow shocks but have investment that is more sensitive to cash flow, suggesting that available internal funds are spent on investment rather than accumulated as cash.